

REG.NO:

2303010621002

GOVERNMENT COLLEGE OF ENGINEERING (AUTONOMOUS), BARGUR

B.E. Degree Examinations – May / June 2025

ELECTRONICS AND COMMUNICATION ENGINEERING

Fourth Semester

22LPC403 –Microprocessor and Microcontroller

Regulations 2022

Time : 3 Hours

Maximum Marks : 100

Answer ALL Questions

Part – A (10 x 2 = 20 Marks)

	M	CO	BT
1. Compare microprocessor and microcontroller.	2	1	AN
2. Write the function of MOVS and CMPS in 8086.	2	1	R
3. Brief about the importance of MN/MX pin in 8086.	2	2	U
4. List the features of multi core processors.	2	2	R
5. State the important functions of an I/O interface.	2	3	R
6. Give the mode instruction and command instruction format of 8251 serial communication interface.	2	3	U
7. Write an assembly language program to add two numbers using 8051.	2	4	AN
8. Write an assembly language program to clear the accumulator and then add '3' to the accumulator ten times using 8051.	2	4	AN
9. Define key debouncing.	2	5	R
10. Give the pin details of 2 × 16 LCD.	2	5	U

Part – B (5 x 16 = 80 Marks)

11. a) (i) Outline in detail about the data transfer and string manipulation instructions with suitable examples.	8	1	U
(ii) Illustrate about modular programming and its advantages.	8	1	U
(OR)			
b) i) Explain in detail about the concepts of procedures.	8	1	U
ii) Illustrate about interrupts and interrupt service routines of 8086.	8	1	U
12. a) Differentiate between minimum and maximum mode of operation of 8086 microprocessor, and sketch the maximum mode configuration.	16	2	AN
(OR)			
b) Differentiate closely coupled and loosely coupled configurations of 8086 based multiprocessing systems, and illustrate the general concepts underlying the schemes for establishing priority in a loosely coupled multiprocessor system.	16	2	AN

13. a) Describe the operating modes and control words of Programmable Peripheral Interface (8255). Also specify the handshaking signals and their functions if port A of 8255 is setup as input port in mode 1. 16 3 R
- (OR)
- b) Describe in detail the interfacing of A/D converter with 8086 with necessary diagram. 16 3 R
14. a) With a functional block diagram, discuss the architecture of the 8051 microcontroller. 16 4 A
- (OR)
- b) Discuss on the different types of addressing modes supported by the 8051 microcontroller with examples. 16 4 A
15. a) i) Illustrate in detail the timers available in 8051 microcontroller. 8 5 A
- ii) Write an assembly language program to control the speed of stepper motor using 8051. 8 5 AN
- (OR)
- b) i) Interface two 4K x 8 EPROMS and two 4K x 8 RAM chips with 8051. Select suitable maps. 8 5 A
- ii) Write an assembly language program to generate a square wave of 50kHz with 50% duty cycle using 8051. 8 5 AN

BT	R	U	A	AN	E	C
%	22	21	27	30	-	-

GOVERNMENT COLLEGE OF ENGINEERING (AUTONOMOUS)

B.E. Degree Examinations – May / June 2025

ELECTRONICS AND COMMUNICATION ENGINEERING

Fourth Semester

22LBS401 - Statistics and Random Processes

Regulations 2022

(Use of statistical table permitted)

Time : 3 Hours

Maximum Marks : 100

Answer ALL Questions

Part - A (10 x 2 = 20 Marks)

- | | M | CO | BT |
|--|---|----|----|
| 1. A continuous random variable X has a probability density function $f(x)=kx$, $0 < x < 1$. Find k . | 2 | 1 | U |
| 2. If the probability that a target is destroyed on any one shot is 0.5, what is the probability that it would be destroyed on 6 th attempt? | 2 | 1 | AN |
| 3. What is a positive and negative correlation? Explain each with an example. | 2 | 2 | R |
| 4. Can $Y=5+2.8X$ and $X=3-0.5Y$ be the estimated regression equations of Y on X and X on Y respectively? Explain your answer. | 2 | 2 | A |
| 5. In the fair coin experiment, we define the process $\{X(t)\}$ as follows.
$X(t) = \begin{cases} \sin \pi t & \text{if head shows} \\ 2t & \text{if tail shows} \end{cases}$. Find $E[X(t)]$. | 2 | 3 | U |
| 6. Given that the autocorrelation function for a stationary ergodic process with no periodic components is $R_{xx}(\tau) = 25 + \frac{4}{1+6\tau^2}$. Find the mean value of the process $\{X(t)\}$. | 2 | 3 | C |
| 7. Prove that the real part of $S_{xy}(\omega)$ is an even function of ω . | 2 | 4 | E |
| 8. Prove that the system $Y(b) = \int_{-\infty}^{\infty} h(\xi)X(t-\xi)d\xi$ is a linear time-invariant system. | 2 | 4 | A |
| 9. What are the mean and variance of t-Distribution? | 2 | 5 | R |
| 10. Define critical region. | 2 | 5 | R |

Part - B (5 x 16 = 80 Marks)

11. a) i) A random variable X has the following probability distribution. 8 1 R

X	-2	-1	0	1	2	3
$p(x)$	0.1	k	0.2	$2k$	0.3	$3k$

- Find k .
- Evaluate $P(-2 < X < 2)$
- Find the cumulative distribution function of X .

- ii) A random variable has the probability density function given by 8 1 A

$$f(x) = \begin{cases} 2e^{-2x}, & x \geq 0 \\ 0, & x < 0 \end{cases} \text{ Find}$$

- the moment generating function
- the first four moments about the origin.

(OR)

- b) i) Find the probability that atmost 5 defective fuses will be found in a box of 200 fuses if experiences shows that 2% of such fuses are defective. 8 1 U
- ii) In a normal distribution, 31% of the items are under 45 and 8% are over 64. Find the mean and variance of the distribution. 8 1 AN

12. a) i) The joint distribution of X and Y is given by $f(x, y) = \frac{x+y}{21}$, $x = 1, 2, 3$ and $y = 1, 2$. Find the marginal distribution of X and marginal distribution of Y . 8 2 R

- ii) Calculate the correlation coefficient for the heights (in inches) of fathers X and their sons Y . 8 2 A

X	65	66	67	67	68	69	70	72
Y	67	68	65	68	72	72	69	71

(OR)

- b) i) The two lines of regression are $8x - 10y + 66 = 0$ and $40x - 18y - 214 = 0$. The variance of x is 9. Find 8 2 U
- a. the mean values of x and y
- b. correlation coefficient between x and y .

- ii) If X and Y each follow an exponential distribution with parameter 1 and are independent, find the pdf of $U = X - Y$, given $f_X(x) = e^{-x}$, $x > 0$ and $f_Y(y) = e^{-y}$, $y > 0$. 8 2 E

13. a) i) If $\{X(t)\}$ is a Gaussian process with $\mu(t) = 10$ and $C(t_1, t_2) = 16e^{-|t_1 - t_2|}$, find the probability that 8 3 R
- a. $X(10) \leq 8$
- b. $|X(10) - X(6)| \leq 4$.

- ii) If customers arrive at a counter in accordance with a poisson process with a mean rate of 2 per minute. Find the probability that the interval between 2 consecutive arrivals is 8 3 AN
- a. more than 1 min
- b. between 1 min and 2 min
- c. 4 min or less.

(OR)

- b) i) Find the mean and variance of the stationary process $\{X(t)\}$, whose auto correlation function is given by $R(\tau) = 2 + 4e^{-2|\tau|}$. 8 3 U

- ii) Find the cross-correlation function of $W(t) = A(t) + B(t)$ and $Z(t) = A(t) - B(t)$ where $A(t)$ and $B(t)$ are statistically independent random variable with zero means and autocorrelation function $R_{AA}(z) = e^{-|\tau|}$, $-\infty < \tau < \infty$ and $R_{BB}(z) = 3e^{-|\tau|}$, $-\infty < \tau < \infty$ respectively. 8 3 R

14. a) i) Find the power spectral density of a wide-sense stationary process with autocorrelation function is $R(\tau) = e^{-\alpha\tau^2}$. 8 4 A

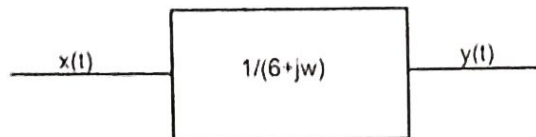
- ii) The cross-power spectrum of real random process $X(t)$ and $Y(t)$ is given 8 4 R

by $S_{xy}(\omega) = \begin{cases} a + \frac{ib\omega}{\alpha}, & -\alpha < \omega < \alpha, \alpha > 0 \\ 0, & \text{otherwise} \end{cases}$. Find the cross-correlation function.

(OR)

- b) i) A wide-sense stationary process $X(t)$ with $R_{xx}(\tau) = Ae^{-a|\tau|}$ where 'A' and 'a' are real positive constants is applied to the i/P of an linear time invariant systems with $h(t) = e^{-bt}u(t)$ where b is a real positive constant. Find the power spectral density of the O/P of the system. 8 4 U

- ii) Consider the linear system as shown below. 8 4 A



$X(t)$ is the input and $Y(t)$ is the output of the system. The auto correlation of $X(t)$ is $R_{xx}(\tau) = 3\delta(\tau)$. Find the power spectral density, auto correlation function and mean square value of the output $Y(t)$.

15. a) i) The fatality rate of typhoid patients is believed to be 17.26 per cent. In a certain year 640 patients suffering from typhoid were treated in a metropolitan hospital and only 63 patients died. Can you consider the hospital efficient? 8 5 U

- ii) Test made on the breaking strength of 10 pieces of a metal wire gave the results: 578, 572, 570, 568, 572, 570, 570, 572, 596 and 584 kg. Test if the mean breaking strength of the wire can be assumed as 577 kg. 8 5 AN

(OR)

- b) i) Two samples of sizes nine and eight gave the sums of squares of deviations from their respective means equal to 160 and 91 respectively. Can they be regarded as drawn from the same normal population? 8 5 AN

- ii) Fit a binomial distribution for the following data and also test the goodness of fit. 8 5 A

x:	0	1	2	3	4	5	6	Total
f:	5	18	28	12	7	6	4	80

BT	R	U	A	AN	E	C
%	25.5	24.4	28.4	19.5	1.1	1.1

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GOVERNMENT COLLEGE OF ENGINEERING (AUTONOMOUS), BARGUR

B.E. Degree Examinations – May / June 2025

ELECTRONICS AND COMMUNICATION ENGINEERING

Fourth Semester

22LPC402 – Linear Integrated Circuits

Regulations 2022

Time : 3 Hours

Maximum Marks : 100

Answer ALL Questions

Part - A (10 x 2 = 20 Marks)

	M	CO	BT
1. List any four ideal characteristics of an op-amp.	2	1	R
2. If an op-amp has a gain-bandwidth product of 2 MHz, find the bandwidth for a gain of 50.	2	1	A
3. Calculate feedback resistance of an operational amplifier with a gain of -5 and input resistance 10KΩ.	2	2	A
4. A precision half-wave rectifier is given a sine wave input with a peak amplitude of 1 V. Sketch both the input and the corresponding output waveforms.	2	2	AN
5. State the function of a high-speed sample and hold circuit.	2	3	U
6. A delta-sigma modulator operates at an oversampling rate of 64 and a sampling frequency of 3.072 MHz. What is the effective bandwidth of the signal?	2	3	A
7. List out the applications of IC 555 timer.	2	4	A
8. In a PLL, if the reference frequency is 10 kHz and the output is locked at 100 kHz, what is the division ratio of the frequency divider?	2	4	A
9. Compare switching regulators and series regulators based on their operation, efficiency, and heat dissipation.	2	5	AN
10. A frequency-to-voltage converter produces an output voltage of 3V for an input frequency of 600 Hz. What is the voltage sensitivity of the converter?	2	5	A

Part – B (5 x 16 = 80 Marks)

11. a) i) Provide a detailed explanation of the general operational amplifier stages and their functions, along with the structure of the IC 741.	10	1	U
ii) List and explain the DC characteristics of an operational amplifier.	06	1	U
(OR)			
b) i) Explain the concept of slew rate in an op-amp, derive its expression, and discuss how it affects the performance of high-frequency signals.	10	1	U
ii) Explain the difference between open-loop and closed-loop configurations in op-amps.	06	1	U
12. a) i) Design a circuit using OP-AMP that will differentiate input signal with $f_{max} = 100 \text{ Hz}$. Draw the output waveforms for a sine wave and a square wave of 1 V peak – to – peak at 100 Hz applied to the circuit.	10	2	A
ii) Explain with a neat circuit diagram of Instrumentation Amplifier and derive the expression for output voltage.	06	2	A

(OR)

- b) i) Design a second order High pass Butterworth filter having high cut off frequency of 1kHz and sketch its frequency response. 10 2 A
- ii) Derive the expression for the log and antilog amplifiers with necessary diagrams. 06 2 A
13. a) i) Design a 4-bit binary weighted resistor DAC for the following specifications: Use LM741 op-amp, $R = 10k\Omega$, $V_{ref} = 2.5V$ and full-scale output = 5V. 10 3 A
- ii) Describe the working principle of a current-steering D/A converter and its applications. 06 3 U

(OR)

- b) i) A dual slope ADC uses a 16-bit counter and 4 MHz clock rate. The maximum input voltage is +10 V. The maximum integrator output voltage should be -8 V when the counter has cycled through 2^n counts. The capacitor used in the integrator is 0.1 μF . Find the value of resistor R of the integrator. 10 3 A
- ii) Discuss the working principle of successive approximation type ADC and its applications. 06 3 U
14. a) i) For an IC 555 timer-based astable circuit, calculate the high state time interval, low state time interval, period, frequency, and duty cycle. 10 4 A
- ii) For the Voltage Controlled Oscillator (VCO) circuit, assume $R_1 = 2.2 K\Omega$, $R_1 = R_3 = 15K\Omega$ and $C_1 = \mu F$. Assume $V_{cc} = 12 V$. Determine the output frequency, the change in output frequency if modulating input V_s is varied from 7V to 8V. 06 4 A

(OR)

- b) i) Construct the block diagram and explain principle of working, characteristics and applications of Frequency synthesizer. 10 4 A
- ii) Determine the output frequency f_o , lock range Δf_L and capture range Δf_c of IC 565 phase locked loop. Assume $R_1 = 15k\Omega$, $C_1 = 0.01\mu F$, $C = 1\mu F$ and the supply voltage is +12 V. 06 4 A
15. a) i) With neat sketch explain the functional block diagram of IC723 general purpose regulator. 10 5 U
- ii) Explain the working of a switched capacitor filter and derive the expression for its equivalent resistance. 06 5 U

(OR)

- b) i) Explain the working principle of a voltage-to-frequency converter. Draw a typical circuit and discuss applications. 10 5 U
- ii) With a neat circuit diagram explain the function of a LM 380 as a power amplifier. 06 5 U

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GOVERNMENT COLLEGE OF ENGINEERING (AUTONOMOUS), BARGUR

B.E. Degree Examinations – May / June 2025

ELECTRONICS AND COMMUNICATION ENGINEERING

Fourth Semester

22LPC404 – Analog Communication

Regulations 2022

Time : 3 Hours

Maximum Marks : 100

Answer ALL Questions

Part - A (10 x 2 = 20 Marks)

	M	CO	BT
1. Compare square law detector with envelope detector.	2	CO1	AN
2. How will you generating DSBSC-AM?	2	CO1	AN
3. Compare the power and bandwidth requirements of AM,FM and PM.	2	CO2	R
4. State the advantages of frequency modulation.	2	CO2	U
5. Formulate the power spectral density of $x(t)$.	2	CO3	C
6. Point out the properties of Gaussian process.	2	CO3	A
7. Find the thermal noise voltage developed across a resistor of 700 ohm. The bandwidth of measuring instrument is 7MHz and the ambient temperature is 27°C.	2	CO4	E
8. What is the purpose of pre-emphasis and de-emphasis in FM?	2	CO4	U
9. Name the three source coding techniques.	2	CO5	R
10. Give the equation for finding the entropy of a binary source.	2	CO5	U

Part - B (5 x 16 = 80 Marks)

11. a) i)	With suitable block diagram and equations, show how will you generate DSB-SC?	08	CO1	U
ii)	Derive the equation of an AM wave. Also draw the modulated AM wave for various modulation index,	08	CO1	AN
(OR)				
b) i)	Derive the expression of amplitude modulation power distribution.	08	CO1	AN
ii)	Explain the detection of AM signals using Envelope detector.	08	CO1	U
12. a) i)	Derive the expression for wide band FM in terms of bessel functions.	10	CO2	AN
ii)	Discuss about the Phased locked loop. What does PLL consist of?	06	CO2	U

(OR)

- b) i) With block diagram explain in detail about Armstrong FM transmitter and its advantages. 10 CO2 R
- ii) Derive the mathematical representation of FM and PM waves. 06 CO2 AN
13. a) i) Explain in detail about the transmission of a random process through a LTI filter. 08 CO3 U
- ii) What is Gaussian random process? State and prove any two properties of Gaussian process. 08 CO3 AN
- (OR)**
- b) i) Write short notes on following terms: mean, correlation, ergodic Process. 08 CO3 R
- ii) Give the properties of the auto correlation function. 08 CO3 U
14. a) i) Derive the noise power spectral density of the FM demodulation and explain its performance with diagram. 12 CO4 AN
- ii) Comment the role of pre-emphasis and de-emphasis circuit in SNR improvement. 04 CO4 AN
- (OR)**
- b) i) Summaries the characteristics of various noise found in a communication channel. 12 CO4 U
- ii) Define white noise. Give its characteristics. 04 CO4 R
15. a) i) Demonstrate the relation between the mutual information and the Joint entropy of the channel input and channel output. 12 CO5 AN
- ii) Write the significance of the entropy $H(X/Y)$ of a communication system where X is the transmitter and Y is the receiver. 04 CO5 AN
- (OR)**
- b) i) Illustrate in detail Huffman coding algorithm and how to encode the source symbols with the set of probabilities using Huffman coding. 12 CO5 AN
- ii) Write a short note on Shannon-Fano coding. 04 CO5 R

BT	R	U	A	AN	E	C
%	15.55	26.67	-	55.56	1.11	1.11

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GOVERNMENT COLLEGE OF ENGINEERING (AUTONOMOUS), BARGUR

B.E. Degree Examinations – May / June 2025

ELECTRONICS AND COMMUNICATION ENGINEERING

Fourth Semester

22LPC405 – Electromagnetic Fields

Regulations 2022

Time : 3 Hours

Maximum Marks : 100

Answer ALL Questions

Part - A (10 x 2 = 20 Marks)

	M	CO	BT
1. Write the Differential Length, surface, volume of Spherical coordinate systems.	2	CO1	R
2. Prove that curl of gradient is zero.	2	CO1	U
3. A uniform line charge, infinite in extent with $\rho_l = 20 \text{ nc/m}$ lies along z axis. Calculate E at (6,8,3)m.	2	CO2	U
4. A Parallel plate capacitor of capacitance $2200 \mu\text{F}$ is connected across a regulated DC power supply of 30V. Calculate the energy stored in the electric field between parallel plates in joules.	2	CO2	U
5. State Gauss's law for magnetic field.	2	CO3	R
6. A circular coil of radius 2m carries a current of 4A. Calculate the magnetic field intensity at the centre.	2	CO3	U
7. List the properties of uniform plane wave.	2	CO4	R
8. Calculate the amplitude of displacement current density inside a capacitor where $\epsilon_r = 6$ & $D = 3 \cos(6 \times 10^6 t) \mathbf{a}_z \text{ C/m}^2$.	2	CO4	U
9. Define skin Effect?	2	CO5	R
10. Define Brewster angle.	2	CO5	R

Part - B (5 x 16 = 80 Marks)

11. a) i) Given $\mathbf{H} = 2.5\mathbf{a}_\theta + 5\mathbf{a}_\phi$ in spherical coordinates. Find Curl of $\mathbf{H}$ at $(2, \pi/6, 0)$.	8	CO1	A
ii) Transform the vector $\mathbf{A} = y\mathbf{a}_x - x\mathbf{a}_y + z\mathbf{a}_z$ in to cylindrical coordinates.	8	CO1	A
(OR)			
b) State and Prove stoke's Theorem and Divergence Theorem.	16	CO1	U
12. a) i) Derive an expression for the capacitance of a pair of coaxial cylinders of radii r_1 and r_2 and length l . the dielectric being air and the outside cylinder is earthed.	10	CO2	A
ii) Determine the capacitance of a parallel plate capacitor with metal plates of size $30\text{cm} \times 30\text{cm}$ separated by 5mm in air medium.	6	CO2	A

- (OR)
- b) Derive the expression of Electric field intensity for Infinite Line of charge. 16 CO2 U
13. a) Derive an expression for magnetic field intensity at any point on axis of circular coil carrying current. 16 CO3 U
- (OR)
- b) i) State Amperes law .Derive Expression for Coaxial cable using amperes law. 10 CO3 U
- ii) Derive expression for scalar and magnetic vector potential. 6 CO3 U
14. a) Derive Maxwell's equation in Point form, Integral form & Phasor form/Harmonically time varying field. 16 CO4 U
- (OR)
- b) i) State Poynting theorem and derive expression for poynting vector. 10 CO4 U
- ii) Derive the expression for Electromagnetic wave equation. 6 CO4 U
15. a) Derive Transmission and Reflection Coefficient for the Plane waves that Incident normally on Dielectric boundary. 16 CO5 U
- (OR)
- b) Derive the expression for plane waves in lossy media. 16 CO5 U

BTL	R	U	A	AN	E	C
%	5.5	85.5	17.7	-	-	-

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GOVERNMENT COLLEGE OF ENGINEERING (AUTONOMOUS), BARGUR

B E. Degree Examinations – May / June 2025

COMMON TO ALL BRANCH

First Semester / Fourth Semester

23PTEBS103 / 22ZBS407 – Environmental Sciences and Sustainability

Regulations 2023 / 2022

Time : 3 Hours

Maximum Marks : 100

Answer ALL Questions

Part – A (10 x 2 = 20 Marks)

	M	CO	BT
1. What is the need for environmental science course in engineering?	2	1	R
2. Differentiate between food chain and food web.	2	1	R
3. What are point and non-point sources of water pollution?	2	2	A
4. What are hazardous and non – hazardous wastes? Give example.	2	2	R
5. What is the need of new energy sources?	2	3	R
6. Mention few applications of hydrogen energy.	2	3	R
7. Define sustainability.	2	4	R
8. What do you mean by carbon foot print?	2	4	U
9. Define circular economy.	2	5	R
10. What do you mean by sustainable transport? Give example.	2	5	AN

Part – B (5 x 16 = 80 Marks)

11. a) i) Explain structure and function of an ecosystem in detail.	8	1	R
ii) Write short notes on ecological succession.	8	1	R
(OR)			
b) i) Explain in detail, the values of biodiversity with examples.	8	1	U
ii) Discuss the In - situ and Ex- situ conservation of biodiversity.	8	1	R
12. a) i) Explain the causes, effects and the suitable preventive measures of air pollution.	8	2	U
ii) Identify the hazardous waste generated by man-made activities. Give its impact on environment.	8	2	A
(OR)			
b) i) Illustrate occupational health and safety management system (OHSMS) with a suitable case study.	8	2	A
ii) Discuss the salient features of Environmental Acts. Explain any two acts in detail.	8	2	R
13. a) Write short notes on	16	3	R
i) Tidal energy			
ii) Geothermal energy.			

(OR)

b) Discuss in detail about environmental benefits and challenges of renewable energy sources such as solar and wind energy. 16 3 R

14. a) Discuss Sustainable Development Goals in detail. Mention any three ways to make a transition from unsustainability to sustainability. 16 4 AN

(OR)

b) Discuss any 2 case studies in environmental management in industry. 16 4 AN

15. a) Briefly explain zero waste and explain the R concept. Discuss impacts of zero waste concept a climate changes. 16 5 U

(OR)

b) Comment on green buildings, Green materials and energy efficiency in creating a sustainable and healthy environment. 16 5 R

BT	R	U	A	AN	E	C
%	20	20	25	35	-	-